

DATA MINING

ACTIVITY 3: CLASSIFICATION

BY GROUP 1:

HARINI

LING

ADRIANA

Class Overview

Today we will be:

Reading chapter 3 of our class text

Completing chapter questions

Learning about data classifications

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Introduction

In this chapter, we dive into the world of data mining, focusing on classification, which is a key technique in supervised learning. Classification involves building models, or classifiers, that can accurately predict category labels for new data by recognizing patterns in data that has already been labeled. We'll compare classification with other prediction methods like regression, outline the general process of classification, and explore some real-world applications, including fraud detection, medical diagnosis, and credit scoring. Additionally, we'll discuss how to evaluate the performance of classifiers using metrics such as accuracy, precision, recall, and the F1-score. The chapter also covers various classification techniques, including decision trees, Bayesian classifiers, neural networks, and support vector machines. To ensure that our classification models are reliable, we'll wrap up by looking at common issues like overfitting and introducing validation strategies such as cross-validation and sampling methods.

Q1 : What is classification in data mining?

A data analysis that extracts models describing important data classes.

how?

builds a model (called a classifier) to maps input data into predefined categories/classes based on historical data.

for example : classify email as “spam” or “not spam”.

Q2 : Explain the importance and real-world applications of classification in data mining.

Real-World Applications :

Importance :

1. Recognize patterns in large datasets
2. Predict outcomes for decision-making
3. Avoid unauthorized access & potential breach

Industry	Example Usage
IT	- Auto-assign support tickets to the correct department. - Detect unusual network behavior in server logs.
Finance	- Detect fraudulent credit card transactions. - Approve or reject loan applications.
Healthcare	- Diagnose diseases based on patient symptoms. - Identify high-risk individuals for routine checks.
Marketing	- Segment customers for targeted campaigns. - Predict which leads are most likely to convert.

What is data classification? (n.d.). Palo Alto Networks.
<https://www.paloaltonetworks.com/cyberpedia/data-classification#:~:text=Data%20classification%20not%20only%20aids,negative%20consequences%20of%20compromised%20security.>

Sharma, R. (2025, April 4). Understanding classification in data mining: types & algorithms, and building a classification model. upGrad Blog. <https://www.upgrad.com/blog/classification-in-data-mining/#What-Are-Some-Real-World-Applications/-Examples-of-Classification-in-Data-Mining/>

**Q3 : Identify at least TEN
(10) classification
algorithms.**

1. **Nearest-Neighbor Classifier**
2. **Bayesian Classifiers**
3. **Rule-Based Classifiers**
4. **Statistical Based Classifiers**
5. **Neural Networks**
6. **Support Vector Machine**
7. **Genetic Algorithm**
8. **Rough Set**
9. **Random Forest**
10. **Logistic Regression**

**Q3 : Then, provide a concise
explanation & example for
THREE (3) of them.**

- **Support Vector Machine**

*Uses the concept of a maximal
margin hyperplane*

e.g. An email filter employs
Support Vector Machines (SVM)
to distinguish between "Spam"
and "Not Spam." It does this by
identifying the optimal boundary
based on the frequency of words
and various features of the
emails.

Q3 : Then, provide a concise explanation & example for THREE (3) of them.

- **Nearest-neighbor Classifier**

Memorizes the entire training data, use distance-based concept

e.g. A health app determines whether a user is "Active" or "Inactive" by analyzing their daily steps and sleep patterns, comparing them to the three users in the dataset who are most similar.

- **Genetic Algorithm**

To classify data samples, nature evolution analogy is used to build directed-search-based mechanism

e.g. A credit scoring system adapts its rules—like considering income and credit history—to categorize users as either "Low Risk" or "High Risk" through a process of selection and mutation.

Q4 : How does classification work? Describe the general steps involved in building a classification model.

1. Perform a risk assessment for sensitive data

to map out and see an infrastructure's vulnerabilities to classify data and prevent future attacks.

2. Establish a data classification policy

to ensure the infrastructure's classification remains accurate.

3. Categorize the types of data

the different levels of classification will determine what needs to be protected.

Q4 : How does classification work? Describe the general steps involved in building a classification model.

4. Identify data locations

to monitor for risks or losing sensitive data.

5. Identify and classify data

categorize the data into low, medium or high sensitivity based on access levels.

6. Use results to improve security and compliance

revisit the data classification policy to check if it's time to update the policy information.

Q5 : Identify and explain common evaluation metrics used in classification in data mining. Include relevant formulas.

Metric	Formula	Evaluation Focus
Accuracy	$ACC = \frac{TP+TN}{TP+TN+FP+FN}$	Overall effectiveness of a classifier
Error rate	$ERR = \frac{FP+FN}{TP+TN+FP+FN}$	Classification error
Precision	$PRC = \frac{TP}{TP+FP}$	Class agreement of the data labels with the positive labels given by the classifier
Sensitivity Specificity	$SNS = \frac{TP}{TP+FN}$ $SPC = \frac{TN}{TN+FP}$	Effectiveness of a classifier to identify positive labels How effectively a classifier identifies negative labels
ROC	$ROC = \frac{\sqrt{SNS^2+SPC^2}}{\sqrt{2}}$	Combined metric based on the Receiver Operating Characteristic (ROC) space [53]
F_1 score	$F_1 = 2 \frac{PRC \cdot SNS}{PRC + SNS}$	Combination of precision (PRC) and sensitivity (SNS) in a single metric
Geometric Mean	$GM = \sqrt{SNS \cdot SPC}$	Combination of sensitivity (SNS) and specificity (SPC) in a single metric

Keta, N. (2023, August 16). Common Evaluation Metrics used for classification tasks — An introduction. Medium.

<https://nachi-keta.medium.com/common-evaluation-metrics-used-for-classification-tasks-an-introduction-2a166bfcd050>

Q6: Compare Supervised Learning and Unsupervised Learning with examples.

SUPERVISED LEARNING

- The training data are accompanied by labels indicating the class of observations
- New data is classified based on the training set
- Example : Classification (Predict whether a patient has cancer)

UNSUPERVISED LEARNING

- The class labels of training data is unknown
- Aim of establishing the existence of classes or clusters in the data
- Example : Clustering (Cluster normal data and detect outliers)

Q7: Differentiate between binary classification and multi-class classification. Provide an example for each.

BINARY CLASSIFICATION

- Data being classified into two classes
- Output labels are only two classes (0 or 1, True or False, Yes or No).
- Example : Spam detection - spam vs. not spam

MULTI-CLASS CLASSIFICATION

- Data being classified into more than two classes
- Output labels are more than two classes
- Example : Plant species classification

Source : <https://www.analyticssteps.com/blogs/binary-and-multiclass-classification-machine-learning>

Q8: Discuss common challenges encountered in classification tasks.

1. **Overfitting** - Occurs when the model is too complex and tries to encode more patterns from the training data than it is necessary. The model ends up hyper-focusing only on patterns causing overfitting.
2. **Underfitting** - Occurs when model is too simple or when the data is noisy or contains complex relationships that cannot be modeled by a simple model, leading to poor performance on both training and testing data.
3. **Class imbalance** - Occurs when one class heavily outnumbered the others, causing the models to be biased toward the majority class.

Q9: What is a confusion matrix and how is it used to evaluate classification models? Provide a labeled example.

	Predicted Positive	Predicted Negative	
Actual Positive	TP <i>True Positive</i>	FN <i>False Negative</i>	Sensitivity $\frac{TP}{(TP + FN)}$
Actual Negative	FP <i>False Positive</i>	TN <i>True Negative</i>	Specificity $\frac{TN}{(TN + FP)}$
	Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

Q9: What is a confusion matrix and how is it used to evaluate classification models? Provide a labeled example.

Confusion matrix is used to:

- **Solve Imbalanced Data:** Using accuracy as the only metric leads to imbalanced datasets. Thus, using other metrics, such as precision and recall through confusion matrix allows a more balanced view and accurate representation of a dataset.
- **Error Type Differentiator:** Confusion matrix reveals all the different type of errors produced by the model.

Source : https://www.analyticssteps.com/blogs/binary-and-multiclass-classification-machine-learning_

Source : https://www.datacamp.com/tutorial/what-is-a-confusion-matrix-in-machine-learning_

Q10: Explain the concept of overfitting and underfitting in classification models. How can each issue be addressed?

OVERFITTING

- Occurs when the model is too complex and tries to encode more patterns from the training data than it is necessary, causing overfitting.
- Also occurs when a small training set, too small to show the patterns in the data ends up hyper-focusing only on patterns.
- The model is not generalizable, which means that the model won't be very good at predicting the targets of data it has never seen before.
- This issue can be tackled by simplifying the model or using cross-validation.

UNDERFITTING

- A situation where a model has yet to learn the true structure of the data.
- The model is too simple or when the data is noisy or contains complex relationships that cannot be modeled by a simple model
- As a results, the model is unable to fit the training data well, leading to poor performance on both training and testing data, as well as high bias and low variance.
- This issue can be tackled by using more complex model

Q11

Describe the role of cross-validation in evaluating the performance of a classification algorithm.

- **Used to evaluate model performance reliably and avoid overfitting.**
 - **Splits data into k folds → each fold used for testing once, others for training.**
 - **Provides unbiased estimate of generalization error.**
 - **Helps in model selection and hyperparameter tuning.**
 - **More accurate than a simple train/test split, especially for small datasets.**
- documentation**

From Slide: Cross-validation ensures the model is tested on unseen data, improving generalization performance.

Reference: [Tan et al., 2018], Scikit-learn documentation

Q12

How can feature selection improve classification model performance? Mention at least two methods

- **Removes irrelevant/redundant features → improves accuracy and training speed.**
- **Helps prevent overfitting and reduces model complexity.**
- **Makes the model easier to interpret and more efficient.**
- **Two Common Methods:**
 - **Filter Method — uses statistical tests (e.g., chi-square, correlation).**
 - **Wrapper Method — uses model performance (e.g., recursive feature elimination).**

From Slide: Reduces errors and helps classifier focus on useful patterns.

Reference: [Guyon & Elisseeff, 2003]

Q13

Compare decision trees and logistic regression in terms of model interpretability, accuracy, and use cases.

Criteria	Decision Tree	Logistic Regression
Interpretability	Easy to understand (if-else rules)	Coefficients explain feature impact (less intuitive)
Accuracy	Good for nonlinear patterns	Better for linear relationships
Overfitting	Prone if not pruned	Less likely due to linearity
Use Case	Rule-based decisions, business logic	Risk scoring, healthcare, social sciences

From Slide: Decision trees popular for explainability; logistic regression used when linear relationship assumed.

Reference: [Scikit-learn docs], Tim Bock – Analytics Comparisons

Q14

Explain what ensemble methods (e.g., Bagging, Boosting, Random Forest) are and how they improve classification performance.

Method	How It Works	Goal	Strengths	Example Algorithm
Bagging	Trains multiple models in parallel on different random bootstrapped subsets	Reduce variance	Prevents overfitting, improves stability	Random Forest
Boosting	Trains models sequentially, each model focuses on previous errors	Reduce bias	Learns from mistakes, improves weak learners	AdaBoost, XGBoost
Random Forest	A form of bagging that also uses random feature selection at each split	Reduce variance, avoid overfitting	Fast, accurate, robust to noise	Random Forest

From Slide: Ensemble methods increase model generalization.
Reference: [Breiman, 1996], Scikit-learn, [Microsoft Docs]

Homework

Due next lesson:

Chapter 3 questions complete

Read to chapter 5



See you
next lesson!